

Exercise C

It is sufficient to show that if $y_i = \alpha + \beta x_i$, then

$$\hat{f}(x) = \sum_{i=1}^m s_i(x) y_i = \alpha + \beta x.$$

This implies that $S\hat{y}_{\text{OLS}} = \hat{y}_{\text{OLS}}$. Note that

$$\hat{f}(x) = \sum_{i=1}^m s_i(x) (\alpha + \beta x_i) = \alpha \sum_{i=1}^m s_i(x) + \beta \sum_{i=1}^m s_i(x) x_i.$$

$$= \alpha \sum_{i=1}^m s_i(x) + \beta \sum_{i=1}^m s_i(x) [x_i - x + x]$$

$$= \alpha \underbrace{\sum_{i=1}^m s_i(x)}_{=1} + \beta \underbrace{\sum_{i=1}^m s_i(x) (x_i - x)}_{=0} + \beta x \underbrace{\sum_{i=1}^m s_i(x)}_{=1} = \alpha + \beta x$$

In fact:

→ This term is not 0 in the Nadaraya-Watson

$$\sum_{i=1}^m s_i(x) = \frac{1}{m} \sum_{i=1}^m \frac{w_i(x) (q_2(x) - (x_i - x) q_1(x))}{q_2(x) q_0(x) - q_1^2(x)} =$$

$$= \frac{1}{m} \frac{\overbrace{\sum_{i=1}^m w_i(x) q_2(x)}^{m q_2(x)} - \overbrace{\sum_{i=1}^m w_i(x) (x_i - x) q_1(x)}^{m q_1(x)}}{q_2(x) q_0(x) - q_1^2(x)}$$

$$= \frac{\cancel{m}}{\cancel{m}} \frac{q_2(x) q_0(x) - q_1^2(x)}{q_2(x) q_0(x) - q_1^2(x)} = 1.$$

Moreover

$$\begin{aligned} \sum_{i=1}^m s_i(x) (x_i - x) &= \frac{1}{m} \frac{\sum_{i=1}^m w_i(x) [q_2(x) - (x_i - x)q_1(x)] (x_i - x)}{q_2(x)q_0(x) - q_1^2(x)} \\ &= \frac{1}{m} \frac{\overbrace{\sum_{i=1}^m w_i(x) (x_i - x) q_2(x)}^{m q_2(x)} - \overbrace{\sum_{i=1}^m w_i(x) (x_i - x)^2 q_1(x)}^{m q_2(x)}}{q_2(x)q_0(x) - q_1^2(x)} \\ &= \frac{1}{m} \cdot \frac{q_1(x)q_2(x) - q_1(x)q_2(x)}{q_2(x)q_0(x) - q_1^2(x)} = 0. \end{aligned}$$

In the Nadaraya-Watson case, we obtain:

$$\begin{aligned} \hat{f}(x) &= \alpha \underbrace{\sum_{i=1}^m s_i(x)}_{\substack{\text{"1"} \\ \text{This is trivially true by definition of NW.}}} + \beta \sum_{i=1}^m s_i(x) (x_i - x) + \beta x \underbrace{\sum_{i=1}^m s_i(x)}_{\substack{\text{"1"} \\ \text{This term is not 0 in the Nadaraya-Watson}}} \\ &= \alpha + \beta x + \underbrace{\sum_{i=1}^m s_i(x) (x_i - x)}_{\text{"Bias term"}} \end{aligned}$$